

CHAPTER 4: RESEARCH DESIGN AND METHODOLOGY

4.1 Introduction

This chapter presents the research design and methodology adopted to develop and evaluate a lightweight Transformer-based forecasting framework for multi-horizon retail demand forecasting under temporally shifted retail demand conditions. Building on the research gap identified in Section 3.5, the study investigates whether knowledge distillation enables a compact Transformer-based student model to retain the forecasting capability of a stronger TFT teacher model while reducing deployment cost.

The study uses the M5 Walmart dataset as the experimental dataset because it provides large-scale daily retail sales data together with calendar information, selling prices, and hierarchical product-store identifiers. The forecasting task is formulated as a 28-day multi-horizon demand forecasting problem, consistent with the M5 forecasting framework.

The proposed methodology is designed around three complementary evaluation dimensions:

1. Forecasting accuracy;
2. Deployment efficiency;
3. Performance under predefined in-distribution (ID) and out-of-distribution (OOD) temporal scenarios.

To align with the research scope, temporal distribution shift is treated as an evaluation concern rather than a drift-detection or adaptation problem. Thus, the proposed forecasting framework is evaluated under predefined ID and OOD temporal scenarios to examine whether the compact student model maintains useful predictive capability when retail demand conditions differ from the training period. The remainder of this chapter first presents the overall research design, followed by the dataset preparation and temporal data partitioning strategy. It then describes the proposed teacher-student forecasting

framework, the knowledge distillation procedure, the experimental setup, and the evaluation methodology.

4.2 Research Design Overview

This study adopts a quantitative experimental research design to investigate the effectiveness of a knowledge-distillation-based lightweight forecasting framework for multi-horizon retail demand forecasting. The research design enables a controlled comparison between forecasting models under identical data preparation, training, and evaluation conditions, allowing the contribution of knowledge distillation to be isolated from that of the compact student architecture itself.

The overall research methodology consists of four main phases:

1. Data preparation, including feature engineering, sequence construction, and chronological data partitioning using the M5 Walmart dataset
2. Teacher model development, where a TFT is trained as the high-capacity reference model.
3. Student model development, where identical compact Transformer student models are trained with and without knowledge distillation to isolate the effect of the distillation strategy.
4. Experimental evaluation, where all models are compared using forecasting accuracy, deployment efficiency, and performance under predefined ID and OOD temporal scenarios.

The comparison between the student model trained with and without knowledge distillation forms the core of the experimental design. Because both student models share the same architecture and differ only in their training strategy, any observed differences in performance can be attributed to the effect of knowledge distillation. The overall research workflow is illustrated in Figure 4.1.

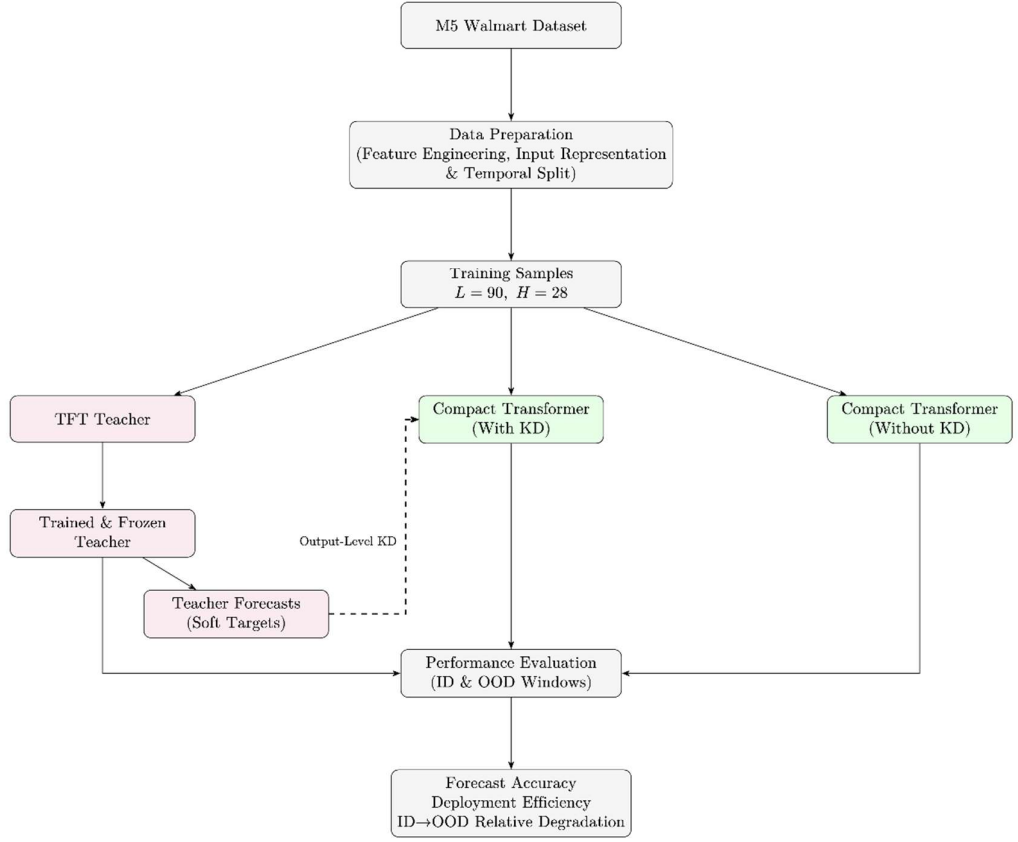


Figure 4.1: Research Design Workflow Overview

To ensure that observed performance differences can be attributed to the proposed modelling approach, all models are evaluated using the same dataset partitions, preprocessing pipeline, input representation, forecasting task formulation, and evaluation protocol. This controlled experimental design allows comparisons between the TFT teacher model, the compact Transformer student without knowledge distillation, and the compact Transformer student with knowledge distillation to focus on differences in model capacity and training strategy rather than differences in data preparation or evaluation conditions.

4.3 Dataset and Data Preparation

4.3.1 M5 Walmart Dataset

This study uses the M5 Walmart retail sales dataset, originally released for the M5 Forecasting Competition. The dataset contains daily sales records for products sold across Walmart stores in the United States and is widely used as a benchmark for retail demand forecasting research. The study primarily uses the `sales_train_evaluation.csv`, `calendar.csv`, and `sell_prices.csv` files, which provide historical sales observations, calendar information, event indicators, SNAP indicators, selling prices, and hierarchical product-store identifiers.

The target variable is daily unit sales. The forecasting task is formulated as a 28-day multi-horizon demand forecasting problem, where future daily sales are predicted for each product-store time series.

The M5 dataset is suitable for the present study because it provides large-scale hierarchical retail sales data together with rich contextual information required for multi-horizon forecasting. Specifically, it supports the following:

- retail demand forecasting;
- multi-horizon forecasting;
- covariate-rich modelling using calendar and price information;
- chronological training and evaluation under predefined temporal scenarios;
- evaluation using established M5 forecasting metrics.

4.3.2 Feature Engineering and Input Representation

The feature engineering process transforms the raw M5 sales, calendar, and price tables into a unified input representation for forecasting. To ensure consistency throughout the experiments, the same feature definitions and preprocessing procedures

are applied throughout the study. The engineered variables are organized into four input groups corresponding to the forecasting input structure adopted by the proposed framework: target series, observed historical inputs, known future inputs, and static categorical inputs. Table 4.1 summarizes the variables included in each input group and their intended purpose.

Table 4.1: Summary of model input groups and variables

Input group	Variables	Purpose
Target series	Daily unit sales	Prediction target for 28-day retail demand forecasting
Observed historical inputs	Historical sales sequence, 7-day lag, 28-day lag, 7-day rolling mean, 7-day rolling standard deviation, zero-sales indicator	Capture recent demand behavior, weekly recurrence, demand variability, and intermittency
Known future inputs	Day of week, month, event name, event type, SNAP indicator, selling price, price-change indicator, percentage price change	Provide calendar and price-related information available at forecast time
Static categorical inputs	Item, department, category, store, and state identifiers	Represent product-store hierarchy and static product/store characteristics

The four input groups define the input representation used by the proposed forecasting framework. The target series consists of the daily unit sales for each product-store time series and forms the prediction target for the 28-day forecasting horizon. Observed historical inputs provide the primary temporal context for forecasting, where the historical sales sequence serves as the primary temporal input, while a small set of engineered features provides complementary information on recent demand patterns and intermittency. Known future inputs provide calendar and price-related information available before the forecasting horizon, enabling the models to utilize future contextual information such as calendar events, SNAP indicators, and selling prices. Static

categorical inputs describe the product-store hierarchy and allow the models to learn persistent differences across products, departments, categories, stores, and states.

Together, these four input groups provide a consistent information set for both the teacher and student models, ensuring that subsequent performance differences reflect differences in model capacity and training strategy rather than differences in feature availability. To ensure fair comparison across all experiments, all historical features are computed exclusively from observations available before the forecasting origin, preventing temporal leakage from future sales values. Numerical variables are standardized using parameters estimated from the training period only, while categorical variables are represented using learnable embeddings. The same preprocessing pipeline, feature definitions, and input schema are applied consistently throughout the study.

4.4 Forecasting Task Formulation

The retail demand forecasting problem is formulated as a supervised multi-horizon forecasting task. For each product-store time series, the model receives a historical observation window of length L and predicts a future demand sequence over a forecasting horizon of length H . The historical input length L denotes the number of past observations provided to the forecasting model before each forecasting origin. L denotes the number of past observations provided to the forecasting model before each forecasting origin, while H denotes the number of future time-steps to be predicted.

In this study, the forecasting horizon is fixed at $H = 28$. This 28-day horizon is consistent with the M5 forecasting setup. Whereas the historical input length is fixed at $L = 90$. The 90-day lookback window is adopted based on the retail forecasting configuration used in the original TFT study, where the Favorita grocery sales dataset was forecast 30 days ahead using 90 days of historical daily observations (Lim et al., 2021). Because the M5 dataset is also a daily retail forecasting dataset with a comparable

28-day forecasting horizon, the same encoder length provides sufficient historical context while maintaining computational efficiency.

For a given forecasting origin t , X_t denotes the model input constructed according to the input representation described in Section 4.3.2. The corresponding prediction target Y_t is the sequence of future daily sales values:

$$Y_t = \{y_{t+1}, y_{t+2}, \dots, y_{t+28}\}$$

Each training sample therefore consists of an input–target pair (X_t, Y_t) . The learning objective is to learn a forecasting function $f(\cdot)$ that maps the model input X_t to the predicted future demand sequence $\hat{Y}_t$, representing the predicted daily demand over the 28-day forecasting horizon:

$$\hat{Y}_t = f(X_t)$$

Training samples are generated by applying a one-day sliding window over the training period. Each training sample consists of a 90-day historical input window and the corresponding 28-day prediction horizon. The window advances one day at a time, allowing all valid input-target pairs within the training period to be used for model training. The final training sample is constrained so that the complete 28-day prediction horizon remains within the training period.

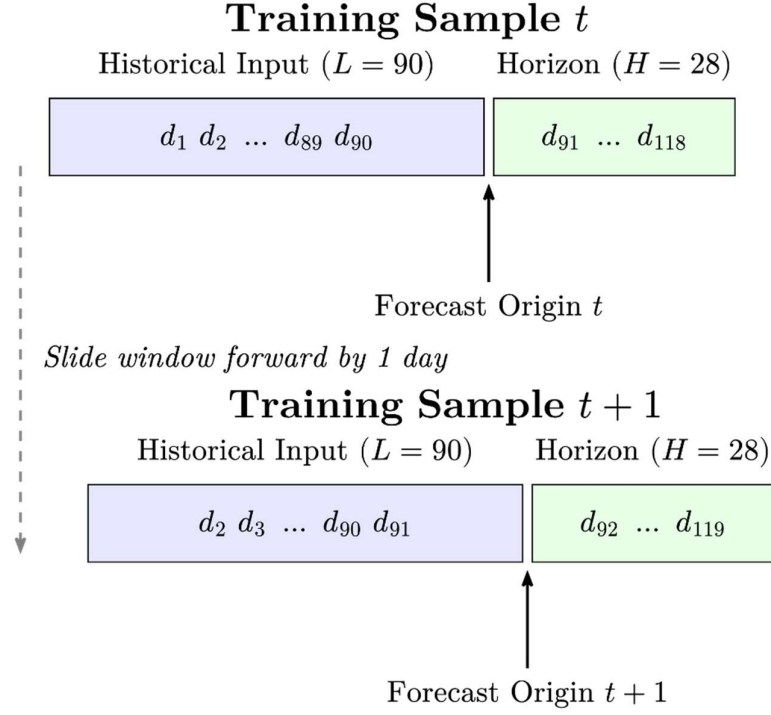


Figure 4.2: Sliding-window sequence construction for $L = 90$ and $H = 28$

The historical input length, forecasting horizon, sliding-window procedure, and data partitions are fixed throughout the study so that all models are trained and evaluated under the same forecasting task.

4.5 Temporal Split and ID/OOD Scenario Design

This study evaluates forecasting performance under both ordinary near-future conditions and a predefined temporally shifted retail scenario. The ID/OOD design is based on chronological splitting of the M5 dataset and uses naturally occurring temporal variation rather than synthetic perturbation. The purpose is not to develop a drift-detection or drift-adaptation mechanism, but to evaluate whether the proposed lightweight model maintains useful forecasting performance under a temporally shifted retail evaluation window.

4.5.1 Chronological Data Split

All temporal splits are defined chronologically to avoid leakage from future periods into model training. Since labelled sales data in `sales_train_evaluation.csv` are available up to `d_1941`, all validation and test windows are defined within this range. The validation, ID, and OOD windows are each fixed for 28 days, consistent with the M5 forecasting horizon. The resulting chronological split is summarized in Table 4.2.

Table 4.2: Chronological split for M5 forecasting experiments

Split	Day range	Calendar range	Duration	Purpose
Training	Day 1 – Day 1,857	2011-01-29 to 2016-02-28	1,857 days	Model training
Validation	Day 1,858 – Day 1,885	2016-02-29 to 2016-03-27	28 days	Hyperparameter tuning
ID test	Day 1,886 – Day 1,913	2016-03-28 to 2016-04-24	28 days	Ordinary near-future evaluation
OOD test	Day 1,914 – Day 1,941	2016-04-25 to 2016-05-22	28 days	Temporally shifted retail evaluation scenario

The ID test window represents an ordinary near-future evaluation period immediately following the validation window, whereas the OOD test window represents a later retail period and serves as the predefined OOD temporal scenario evaluated in this study. Both evaluation windows are defined before model development and used consistently throughout all forecasting experiments.

4.5.2 Justification of the ID and OOD Evaluation Windows

To justify the predefined ID and OOD evaluation windows, a focused exploratory analysis was conducted before model development using calendar, sales, price, and distributional indicators derived from the M5 dataset. The purpose of this analysis is to

characterize the predefined windows rather than to select evaluation periods based on model performance. The complete EDA, including extended descriptive statistics and visualizations, is provided in [Appendix A](#). Table 4.3 summarizes the main indicators used to compare the ID and OOD windows.

Table 4.3: Key characteristics of predefined ID and OOD evaluation windows

Category	Metric	ID test	OOD test
Demand	Zero-sales ratio	0.5628	0.5444
Demand	Active SKU ratio	0.9480	0.9734
Event	Event frequency	0.0000	0.1429
Event	Holiday count	0	2
Price	Average SKU price CV	0.0011	0.0016
Price	Price-change frequency	0.0008	0.0010
Distribution	KS statistic vs training	0.5042	0.5839
Distribution	Wasserstein distance vs training	8,165.89	9,885.10

Figure 4.3 provides a visual comparison of selected characteristics between the predefined ID and OOD windows.

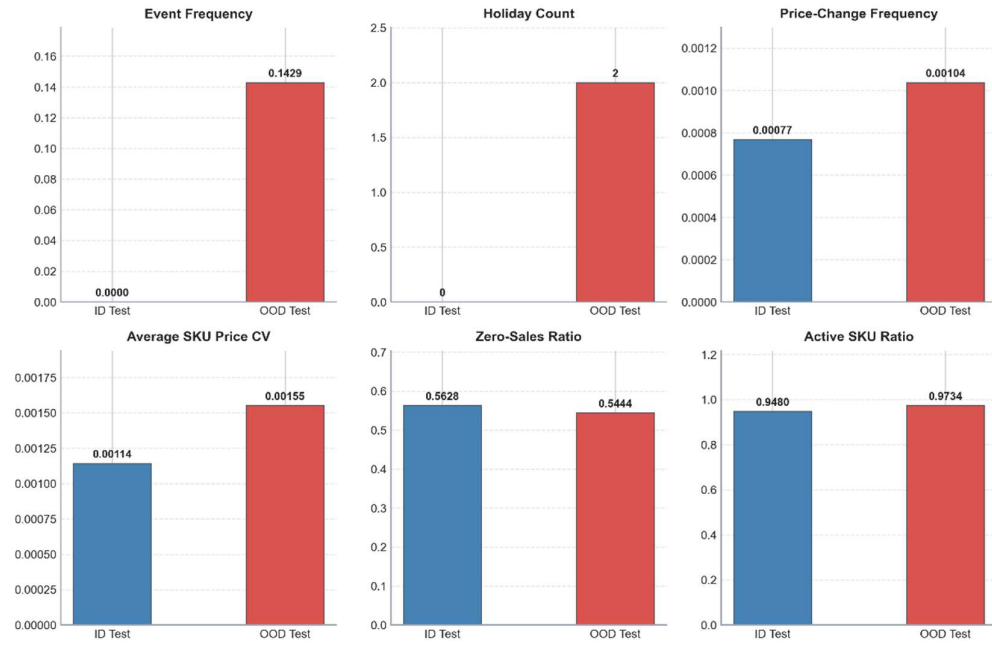


Figure 4.3: Comparison of selected ID and OOD window characteristics

Together, Table 4.3 and Figure 4.3 show that the OOD window differs from the ID window across several retail-relevant indicators. The OOD window has higher event exposure, greater price-change activity, higher active SKU ratio, lower zero-sales ratio, and larger aggregate distributional distance from the training period. Therefore, the OOD window is treated as a predefined event-intensive temporally shifted retail evaluation scenario. This scenario is not interpreted as a severe synthetic distribution shift, but as a realistic temporal retail condition that differs from the ordinary ID test window.

4.5.3 ID/OOD Performance Evaluation

For each forecasting model, forecasting performance will be evaluated independently on the ID and OOD test windows. Relative degradation will then be computed to quantify the percentage increase in forecasting error when moving from the ordinary ID evaluation window to the predefined OOD temporal scenario.

For an error metric where lower values indicate better performance, relative degradation is defined as:

$$\text{Relative Degradation (\%)} = \frac{\text{OOD Error} - \text{ID Error}}{\text{ID Error}} \times 100$$

A lower relative degradation indicates that a model maintains more stable forecasting performance across the ID and OOD evaluation windows.

Relative degradation will be computed and reported for the TFT teacher model, the compact Transformer student without knowledge distillation, the compact Transformer student with knowledge distillation, and the seasonal naïve baseline. This enables comparison of whether knowledge distillation helps the compact Transformer model preserve forecasting performance under temporally shifted retail conditions.

4.6 Proposed Teacher-Student Forecasting Framework

4.6.1 Framework Overview

As shown in Figure 4.1, the proposed forecasting framework adopts a teacher-student knowledge distillation design for lightweight multi-horizon retail demand forecasting. The framework consists of three forecasting models: a high-capacity TFT teacher model, a compact Transformer student model trained without knowledge distillation, and the same compact Transformer student model trained with knowledge distillation.

The TFT teacher model is trained first using the chronological training samples generated from the M5 dataset. After training, the teacher model is frozen and used to generate output-level multi-horizon forecasts. These teacher forecasts serve as soft targets for the compact Transformer student trained with knowledge distillation. In contrast, the student without knowledge distillation is trained using only the ground-truth demand labels. This design allows the study to isolate the contribution of knowledge distillation.

Since both student variants share the same architecture, input representation, forecasting task, and experimental settings, any performance difference can be attributed primarily to teacher-guided learning rather than differences in model design or data preparation.

4.6.2 TFT Teacher Model

Consistent with the literature reviewed in Chapter 3, the TFT is adopted as the teacher model because it provides a strong forecasting baseline for covariate-rich multi-horizon retail demand forecasting. In this study, the TFT teacher receives the shared input representation described in Section 4.3.2 and learns the supervised forecasting task formulated in Section 4.4.

The TFT teacher serves as a high-capacity forecasting reference rather than a lightweight deployment model. After training and validation, the teacher parameters are frozen, and the trained teacher is used to generate output-level forecasts for student distillation. The teacher model is also evaluated directly on the predefined ID and OOD test windows to provide a reference point for forecasting accuracy, deployment efficiency, and ID - OOD relative degradation.

4.6.3 Compact Transformer Student Model

The student model is designed as a compact Transformer-based forecasting model. Since the study focuses on lightweight Transformer models, the student retains an attention-based temporal modelling structure but reduces representational capacity compared with the TFT teacher. The compact student acts as a smaller deployable approximation of the teacher model, allowing the study to examine whether knowledge distillation can help preserve forecasting performance while reducing deployment cost.

The compact Transformer student uses the shared input representation described in Section 4.3.2. These inputs are transformed into a compact representation and processed

by a lightweight Transformer encoder. The student then produces a direct 28-day multi-horizon demand forecast, consistent with the forecasting task defined in Section 4.4.

Table 4.4: Summary of compact Transformer student design

Component	Design
Model type	Compact Transformer-based forecasting model
Input representation	Shared input representation from Section 4.3.2
Temporal backbone	Lightweight Transformer encoder
Model capacity	Reduced compared with TFT teacher
Tunable configuration	Encoder layers, hidden dimension, attention heads, feed-forward dimension, dropout rate, learning rate
Output	Direct 28-day multi-horizon forecast
Role	Lightweight student for evaluating the effect of knowledge distillation

The final student configuration is selected through validation-based hyperparameter tuning and fixed before the final experiments. Once selected, the same student configuration is used for both the student-without-KD and student-with-KD experiments. This ensures that the effect of knowledge distillation can be evaluated independently of changes in student architecture.

4.6.4 Output-Level Knowledge Distillation Strategy

This study applies output-level knowledge distillation. The compact Transformer student is trained to learn from both the ground-truth demand values and the forecasts generated by the trained and frozen TFT teacher model. The teacher’s final multi-horizon forecasts are used as soft targets for the student with KD. The total student training loss is defined as:

$$L_{total} = \alpha L_{supervised} + (1 - \alpha) L_{distillation}$$

where $L_{supervised}$ denotes the supervised forecasting loss between the student forecast and the ground-truth demand values; $L_{distillation}$ denotes the distillation loss between the student forecast and the TFT teacher forecast; and α controls the balance between direct supervision and teacher-guided learning. The weighting coefficient α is treated as a tunable hyperparameter and selected using the validation window.

Only output-level knowledge distillation is considered. Hidden-state, attention-map, and variable-selection distillation are excluded because they require additional architectural alignment between the teacher and student models. This keeps the methodology focused on whether teacher-generated forecasts can help a lightweight Transformer model preserve forecasting performance under deployment and temporal-shift evaluation constraints.

The supervised component trains the student to predict the actual demand values, while the distillation component encourages the student to approximate the teacher model’s multi-horizon forecasting behavior. This allows the compact Transformer student to learn not only from observed labels, but also from the predictive patterns captured by the stronger TFT teacher.

4.6.5 Model Development Workflow

The model development workflow follows the teacher-student framework shown in Figure 4.1. The TFT teacher is first trained using the chronological training data and validated using the validation window. After training, the teacher parameters are frozen, and teacher forecasts are generated for student training.

Two compact Transformer students are then trained using the same architecture and training samples. The first student is trained using only ground-truth demand labels, while

the second additionally incorporates teacher forecasts through the output-level knowledge distillation objective defined in Section 4.6.4. Once training is completed, the TFT teacher model and both compact Transformer student variants are evaluated on the predefined ID and OOD test windows. This completes the model development process and prepares the models for the comparative evaluation described in the following section.

4.7 Baseline Models and Comparative Setup

This study evaluates the proposed teacher-student forecasting framework by comparing the compact Transformer models with representative forecasting baselines. The comparison is designed to isolate the contribution of knowledge distillation, quantify the trade-off between forecasting accuracy and deployment efficiency, and assess model performance under predefined ID and OOD temporal scenarios. Table 4.5 summarizes the models included in the experimental comparison.

Table 4.5: Model comparison setup

Model	Role in experiment	Purpose
Seasonal naïve	Statistical baseline	Provides a simple benchmark for verifying forecasting performance
TFT teacher	High-capacity reference model	Serves as the teacher model and upper-performance reference
Compact Transformer student without KD	Ablation baseline	Quantify the performance of compact architecture without teacher guidance
Compact Transformer student with KD	Proposed model	Evaluates the effectiveness of knowledge distillation for lightweight forecasting

Four principal comparisons are conducted. First, the compact Transformer student trained with knowledge distillation is compared with the same student architecture trained

without knowledge distillation to isolate the contribution of teacher-guided learning. Second, the student with knowledge distillation is compared with the TFT teacher model to assess the extent to which forecasting performance is retained while improving deployment efficiency. Third, all learning-based models are evaluated on both the ID and OOD test windows to examine performance stability under temporally shifted retail conditions. Finally, the seasonal naïve model provides a simple statistical benchmark against which the forecasting capability of the learning-based models can be assessed.

The objective of the experimental comparison is not to demonstrate superiority over every possible forecasting model, but to evaluate whether knowledge distillation enables a compact Transformer model to achieve a favorable balance between forecasting accuracy, deployment efficiency, and performance stability under temporally shifted retail conditions.

4.8 Hyperparameter Tuning and Model Selection

Hyperparameter tuning is performed using the validation window only. The predefined ID and OOD test windows are reserved exclusively for final model evaluation and are not used during model selection, thereby preventing information leakage into the evaluation process.

For the compact Transformer student, validation-based hyperparameter tuning is used to determine the final model configuration. The tuned hyperparameters include the number of encoder layers, hidden dimension, number of attention heads, feed-forward dimension, dropout rate, learning rate, and the distillation weighting coefficient, α . The same final student configuration is used for both the student without knowledge distillation and the student with knowledge distillation, with the latter additionally incorporating the distillation objective during training. Once selected, the final

hyperparameter configuration is fixed and used consistently for all subsequent experiments.

The final student model is selected based on a balance between validation forecasting performance and deployment efficiency. Deployment efficiency is assessed using the number of trainable parameters, model size, and inference time. This procedure ensures that the compact Transformer model is selected systematically rather than arbitrarily while remaining aligned with the study’s objective of achieving a balance between forecasting accuracy and lightweight deployment.

4.9 Evaluation Metrics

The models are evaluated across three main dimensions: forecasting accuracy, deployment efficiency, and temporal stability under predefined ID and OOD temporal scenarios. Table 4.6 summarizes the evaluation metrics used in this study.

Table 4.6: Evaluation metrics and purpose

Evaluation aspect	Metric	Purpose
Forecasting accuracy	WRMSSE, MAE, RMSE, MASE, WAPE	Assess forecasting performance across the 28-day prediction horizon
Deployment efficiency	Trainable parameter count, saved model size, inference time	Assess lightweight deployment advantage
Temporal stability	Relative ID-to-OOD degradation	Assess performance stability under temporally shifted retail conditions

For all forecasting error metrics, lower values indicate better predictive performance. Similarly, lower relative ID-to-OOD degradation indicates more stable forecasting performance under the predefined OOD temporal scenario.

Forecasting accuracy is evaluated primarily using WRMSSE, which is aligned with the M5 forecasting evaluation setting. MAE, RMSE, MASE, and WAPE are also reported to provide complementary measures of forecasting error and interpretability. Forecasting performance is reported for the overall 28-day prediction horizon and, where applicable, for short-, medium-, and long-horizon prediction ranges, as well as separately for the predefined ID and OOD test windows.

Deployment efficiency is evaluated using the number of trainable parameters, model size, and inference time. These metrics assess whether the compact Transformer student provides practical deployment advantages over the TFT teacher model. The lightweight claim is therefore based on model compactness and prediction-time efficiency rather than training duration.

Temporal stability is assessed using the relative ID-to-OOD degradation defined in Section 4.5.3. This metric measures the percentage increase in forecasting error when moving from the ID test window to the predefined OOD temporal scenario. Lower relative degradation indicates that a model maintains more stable forecasting performance under temporally shifted retail conditions.

4.10 Experimental Procedure

The overall experimental procedure follows the teacher–student forecasting framework shown in Figure 4.1. The following steps summarize the end-to-end workflow used to develop, train, and evaluate the forecasting models.

1. Load the M5 sales, calendar, and price datasets.
2. Construct the predefined chronological training, validation, ID, and OOD data partitions.

3. Generate the shared input representation and construct supervised forecasting samples using a 90-day historical input window and a 28-day prediction horizon.
4. Train the TFT teacher model using the training data and tune its hyperparameters using the validation window.
5. Freeze the trained TFT teacher model and generate teacher forecasts as soft targets for knowledge distillation.
6. Train the compact Transformer student without knowledge distillation using only the ground-truth demand labels.
7. Train the compact Transformer student with knowledge distillation using both the ground-truth demand labels and the teacher-generated forecasts.
8. Evaluate the TFT teacher model and both compact Transformer student variants on the predefined ID and OOD test windows.
9. Compute forecasting accuracy, deployment efficiency, and relative ID-to-OOD degradation for each model.
10. Compare the performance of all models based on the predefined evaluation metrics.

These steps summarize the experimental workflow adopted throughout the study and provide a consistent procedure for model development, evaluation, and comparison.

4.11 Summary

This chapter presented the research methodology for evaluating a knowledge distillation (KD)-based lightweight Transformer framework for multi-horizon retail demand forecasting using the M5 Walmart dataset. The proposed methodology comprises data preparation, feature engineering, input representation, forecasting task formulation,

temporal split design, teacher-student model development, baseline comparison, hyperparameter tuning, evaluation metrics, and the overall experimental procedure.

A chronological experimental design with predefined training, validation, ID, and OOD evaluation windows was established to support consistent model development and assessment. The OOD window represents a temporally shifted retail evaluation scenario, enabling the study to examine forecasting performance under both ordinary and shifted retail conditions.

The proposed framework will be evaluated by comparing the TFT teacher model, the compact Transformer student without knowledge distillation, and the compact Transformer student with knowledge distillation using forecasting accuracy, deployment efficiency, and temporal stability as the primary evaluation criteria. The methodology presented in this chapter provides the foundation for the experimental implementation and performance analysis described in the following chapter.